

Brightness Setting	Switch State	Expected Lamp Brightness	Observed Output
0	On	0	0
50	On	50	50
100	On	100	100
0	Off	0	0
50	Off	0	0
100	Off	0	0

Function Parameters and Return Values

bool switchOn: Accepts a true or false value used to represent the state of the lamp, whether it is switched on (true) or off (false).

int brightnessSetting: Accepts an integer value (0 - 100) that is used to represent the desired lamp's brightness.

Return Value: Returns an integer value representing the actual brightness of the lamp. It returns the brightnessSetting if switchOn is true, and returns 0 if switchOn is false.

Coding Check

1. In `int brightnessSetting = 75;`, identify the data type, variable name, and initial value. How would you change the value to 50?

Data type is integer (int), variable name is brightnessSetting, initial value is 75, and to change the value to 50, we need to change the assigned value from 75 to 50.

2. Why is bool suitable for switchOn, while int is suitable for brightnessSetting?

It is suitable to use bool for switchOn since we only need two states for it, which is on and off thus making use of bool's true and false data type suitable for it. While we use int for brightnessSetting since we need more than just two values, and the "variability" of int in which you could use all the integers as values (in this context, 0 - 100) makes it suitable to be used in this variable.

3. Predict the output when brightnessSetting is 80 and switchOn is false.

The observable output will be lamp brightness = 0, since despite using 80 as the value for brightnessSetting, one of the conditions of the return value to be 0 is when switchOn is false, thus giving us the output value of 0.

4. Are switchOn and switchon the same variable? Explain.

No they are not the same variable, since variable names in C++ are case sensitive, so a single difference of uppercase and lowercase characters make things different.

5. What values are passed to getLampBrightness(), and what does it return?

The values passed are the bool value of switchOn and int value of brightnessSetting, the return value will then be the value of brightnessSetting that was passed if switchOn is true, else if switchOn is false, the returned value will be 0.

6. If lampBrightness is declared inside setup(), can loop() access it directly? Explain.

No, variables that are declared in setup() only have local scope and usage within setup(), loop() can not access them unless they are inside loop(), or declared as a global variable, or are used and/or passed as arguments.